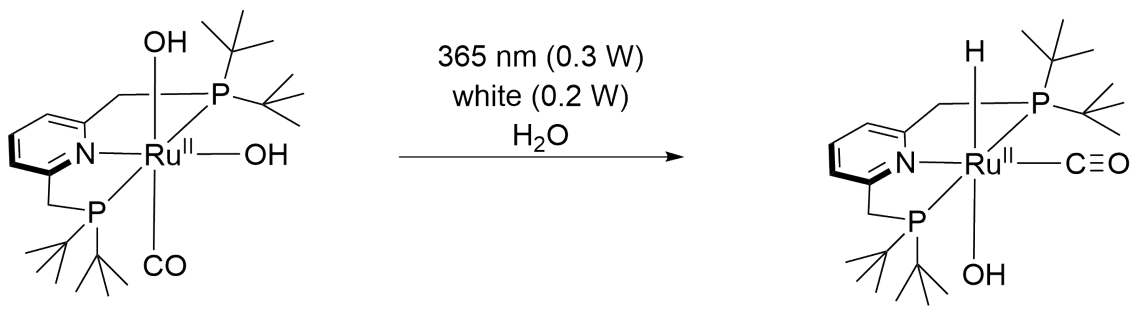
AE-358: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, 50 mM KOH

Date: 2024-10-16
Tags: Dihydroxo Radiation PNP P(tBu)N(py)P(tBu) [Ru(CO)(OH)2(PNP)] NMR AE 1H 31P
Category: Two-photon
Status: Done
Created by: Alexander Eith

Reaction scheme/sample structure

Radiation of Dihydroxo



Chemical Formula: $C_{24}H_{45}NO_3P_2Ru$
Molecular Weight: 558,64552
[Ru(CO)(OH)2(PNP^{tBu})]

Chemical Formula: $C_{24}H_{45}NO_2P_2Ru$
Molecular Weight: 542,64652
[Ru(CO)H(OH)(PNP^{tBu})]

Literature/reference experiments

Literature	Procedure after https://doi.org/10.1039/D1EE01053K
Reproduction	/
Related experiment	AE-348: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, water, longer irrad Two-photon - AE-357: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, 1000 mM KOH

Reagents

Name	CAS/Experiment number	Amount [µmol]	Equivalents	Mass$_{theo}$ [mg]	Mass$_{exp}$ [mg]	Molar mass [g/mol]	Volume [ml]
[Ru(CO)(OH)2(PNP{tBu})]	AE-310: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h (**AE-310-1**)	4.55	/	/	2.54	558.65	/
Milli-Q water	KRA-104: Degassing of Milli-Q water	/	/	/	/	/	0.855
KOH (1M)	AE-243: 1 M Stocksolution of KOH in water	/	/	/	/	/	0.045

Irradiation Parameters

Power measured using [Power Meter] 843-R-USB + 919P-020-12 unless specified otherwise.

	Light Source Name	Wavelength [nm]	Power Setting [mW]
First light source	Light Source - LCS-0365-76-22	365 nm	300
Second light source	Light Source - LCS-6500-65-22	white	200

Used beam combiner [Name or None]	Beam Combiner - LCS-BC25-0409
Irradiation distance [cm]	
Thermostat temperature [°C]	25

Procedure/observations

All steps were carried out using [Protocol] Schlenk Technique. All steps were carried out in the dark, unless other specified.

Date	Time	Step	Observations
15.10		A 10 mL flask was prepared	
	10:10	To the flask [Ru] was added under air	A small amount of the substance was lossed during the transfer
		The flask was set under argon	
	15:00	To the flask water was added	
	15:00	To the flask KOH (1M) was added	A 50 mM KOH solution was obtained
	15:05	The obtained mixture was stirred at 320 rpm.	AE-358 right.jpg
	15:35	From the obtained solution an NMR sample was prepared according to Protocol - Preparation of NMR Sample using a D2O/H3PO4 capillary. (**AE-358-1**) During the transfer the sample was filtered through a syringe filter (PA, pore diameter = 0.2 µm)	AE-358-1.jpg
17.10	12:15	The setup from Two-photon - AE-357: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, 1000 mM KOH was used without any modification	
	12:16	The NMR-tube was placed in the setup accroding to [Protocol] Irradiation setup	
	12:17 - 15:17	The J.Young NMR tube was irradiated for 3:00 h	irrad setup.jpg

	15:29	An NMR was measured (**AE-358-2**)	AE-358-2.jpg
22.10	10:00	An NMR was measured (**AE-358-3**)	AE-358-3.jpg

Analysis

Date	Time	Sample name	Analysis Method	Used Device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
15.10	21:21	AE-358-1	31P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-358-1.zip	AE-358-1_10.nmrium	/	Looks as expected
21.10	10:01	AE-358-2	31P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-358-2.zip	AE-358-1_10.nmrium	/	Two 31P NMR where measured, both look identical and where maesured the same way. For evalutaion -10 was used.
22.10	12:01	AE-358-3	31P {1H}quant, 1Hzgespg, 1H-DOSY	IAAC 400 MHz I	H2O	AE-358-3.zip	/	/	Only the water signal is observed in the DOSY. Otherwise nothing unexpected.

Results

Sample	Yield (hydride species) [%]	Yield (dihydroxo) [%]	Data
AE-358-2	19	7	AE-358-2_Yield_Determination.xlsx

Good agreement with result after 3 h from Two-photon - AE-349: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, 50 mM KOH, shorter irrad (AE-349-1-2)

For comparison with other experiments see: Two-photon - AE-371: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, 0.1 mM KOH

Linked experiments

- AEI-022: Radiation of [Ru(CO)(OH)2(PNN)], NO2-route, 365 nm and white LED
- AEI-026: Capillary production
- AEI-064: Synthesis of [Ru(CO)(OH)2(PNP)] with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag2O

- AEI-067: Radiation of [Ru(CO)(OH)2(PNP{tBu})], crude, 365 nm and white

- AEI-071: Radiation of [Ru(CO)(OH)2(PNP{tBu})], purer, 365 nm and white

- AE-105: Synthesis of [Ru(CO)(OH)2(PNP)] with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag2O

- AE-110: Radiation of [Ru(CO)(OH)2(PNP{tBu})], purified, 365 nm and white

- AE-119: Radiation of [Ru(CO)(OH)2(PNP{tBu})], purified, 52 mM KOH, 365 nm and white

- AE-140: Radiation of [Ru(CO)(OH)2(PNP{tBu})], purified, 365 nm and white, longer duration I

- AE-145: Synthesis of [Ru(CO)(OH)2(PNP)] with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag2O

- AE-158: Radiation of [Ru(CO)(OH)2(PNP{tBu})], purified, 365 nm and white

- KRA-092: Radiation of [Ru(CO)(OH)2(PNPtBu)], 365 nm and white

- KRA-104: Degassing of Milli-Q water

- AE-238: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h

- AE-243: 1 M Stocksolution of KOH in water

- AE-246: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-238, 97 % pure) 365 nm and white

- AE-310: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h

Two-photon - AE-348: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, water, longer irrad

Two-photon - AE-349: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, 50 mM KOH, shorter irrad

Two-photon - AE-357: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, 1000 mM KOH

Two-photon - AE-371: Radiation of [Ru(CO)(OH)2(PNP{tBu})] (AE-310) 365 nm and white, 0.1 mM KOH

Linked resources

Beam Combiner - Beam Combiner LCS-BC25-0409 -1

Light Source - UHP LED 365 nm-1

Light Source - UHP LED 6500 K (white)-1

Protocol - Preparation of NMR Sample

Protocol - Schlenk Technique

Protocol - Irradiation setup

Attached files

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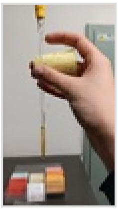

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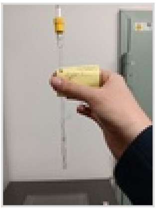

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Link: https://elab.water-splitting.org/experiments.php?mode=view&id=1421